Computer

Vision

CO1-AT2-Concept Mapping

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CO1-AT2 Concept Mapping

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1. Overview & Features

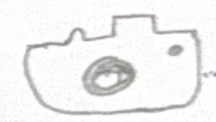
- Enables machines to understand & interpret visual data
- Detects, recognises & tracks the object
- Understands environment

2. Digital Image Fundamentals

- Image is 2D array pixels
- Pixel has an intensity value
- Types: grayscale & color images
- Resolution

3. Image Sensing & Acquisition

- Sensors: cameras, HD AR, ultrasonic, infronted
- Capture real-world visual information from the environment
- Convert light signals to electrical signals



4. Sampling & Quantization

- Spatial Sampling: Converts a continuous image into a grid of pixels



- Intensity quantization: Converts continuous intensity values into discrete values



Autonomous Vehicle

5. Image formation Models

- Illumination: light from the env
- Reflection: Object reflect light to camera
- Projection: 3D world mapped to 2D image using camera model
- Geometric model: Relates 3D points

7. Applications

- Lane detection & lane keeping.
- Pedstrian & cyclist detection
- Navigation & path planning
- Self driving design making.

2.

1. Overview & feature

- Helps in automatic analysis of medical image
- Detects & recognize anatomical structure
- Improves accuracy & reduces human error

2. Digital Image Fundamentals

- Image is 2D array of pixels
- Each pixel has a intensity value
- Types: Gray scale & color images
- Resolution (Image size & clarity)

3. Image Sensing & Acquisition

- X-ray transmit through the body
- X-ray detector
- Converts x-ray energy into electrical signals.

4. Sampling & Quantization

- Sampling Converts the continuous x-ray image into grid of pixels
- Determines the image resolution.
- Forms a digital image suitable for processing & analysis

Medical Imaging

Applications

- Detection of abnormalities
- Classification of disease
- Segmentation of organs & tissues
- Measurement
- Treatment planning & monitoring

5. Image Formation Models

- X-ray projection model: 3D body structure projected to 2D image
- $I(x,y) = I_0 e^{-\mu t}$
  - I_0 incident x-ray
  - μ : attenuation co-efficient
  - t : thickness & tissue.